

Classifying Penguin Sex Using Decision Trees (Palmer Penguins Dataset)

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Define the problem

Our project aims to answer this question: **can we predict a penguin's sex based on its physical characteristics, and which traits are the strongest indicators of sex?** We decided to use the data on penguins and chose this question because we are interested in how real and measurable traits can be used to highlight or predict patterns in nature. Penguins are an interesting subject because they have several physical, measurable characteristics, like body mass, beak size, and flipper length, which differ between sex and species.

Data collection

The dataset that we're using for this project is called "Palmer Penguins" and it was originally collected by Dr. Kristen Gorman and the Palmer Station, Antarctica LTER (Long-Term Ecological Research) program. It was collected and used for a paper titled "Ecological Sexual Dimorphism and Environmental Variability within a Community of Antarctic Penguins (Genus *Pygoscelis*)" by Kristen B. Gorman, Tony D. Williams, and William R. Fraser. It was published on March 5, 2014 in the PLOS One peer-reviewed journal. The population of this dataset is the penguins living in the Palmer Archipelago (Biscoe, Dream, and Torgersen islands) near the Palmer Station, Antarctica, specifically from the Adelie, Gentoo, and Chinstrap species observed between 2007 and 2009. There are 344 samples (each individual penguin), and there are 17 features. The features consist of:

- Study name
- Sample number
- Species
- Region
- Island
- Stage
- Individual ID
- Clutch Competition
- Date Egg
- Culmen Length (mm)
- Culmen Depth (mm)
- Flipper Length (mm)
- Body Mass (g)
- Sex
- Delta 15 N (o/oo)
- Delta 13 C (o/oo)
- Comments

When it comes to potential limitations, there seem to be some missing values, especially in the sex column, which could limit how much of the data we can actually use. Also, the data is only from penguins from one region in Antarctica, so it might not generalize to penguin populations in other parts of the continent.

Data preparation

Before we actually analyzed the data, it was important that we cleaned and organized the dataset so that we obtained accurate results. First, we dropped all columns except the columns needed for the machine learning model, as the other columns weren't needed for the train and test sets. To do this, we created a new dataset that was a subset of the original dataset but only contained the "Sex", "Culmen Length (mm)", "Culmen Depth (mm)", "Flipper Length (mm)", and "Body Mass (g)" columns. Next, we handled the null values that were in several of the rows, and especially in the sex column as that was our target variable. We used the pandas function `dropna()` to ensure that there weren't any missing values in the rows we analyzed. Finally, we needed to turn the categorical values in the "Sex" column into numerical values (1s and 0s). This is because we used functions from scikit-learn, which don't know how to handle text values like "MALE" or "FEMALE", we had to turn them into numbers. We did this by assigning the value of 1 for each "FEMALE" in the column and the value of 0 for each "MALE" in the column. Once these steps were complete we had a clean dataset ready for analysis.

Data exploration

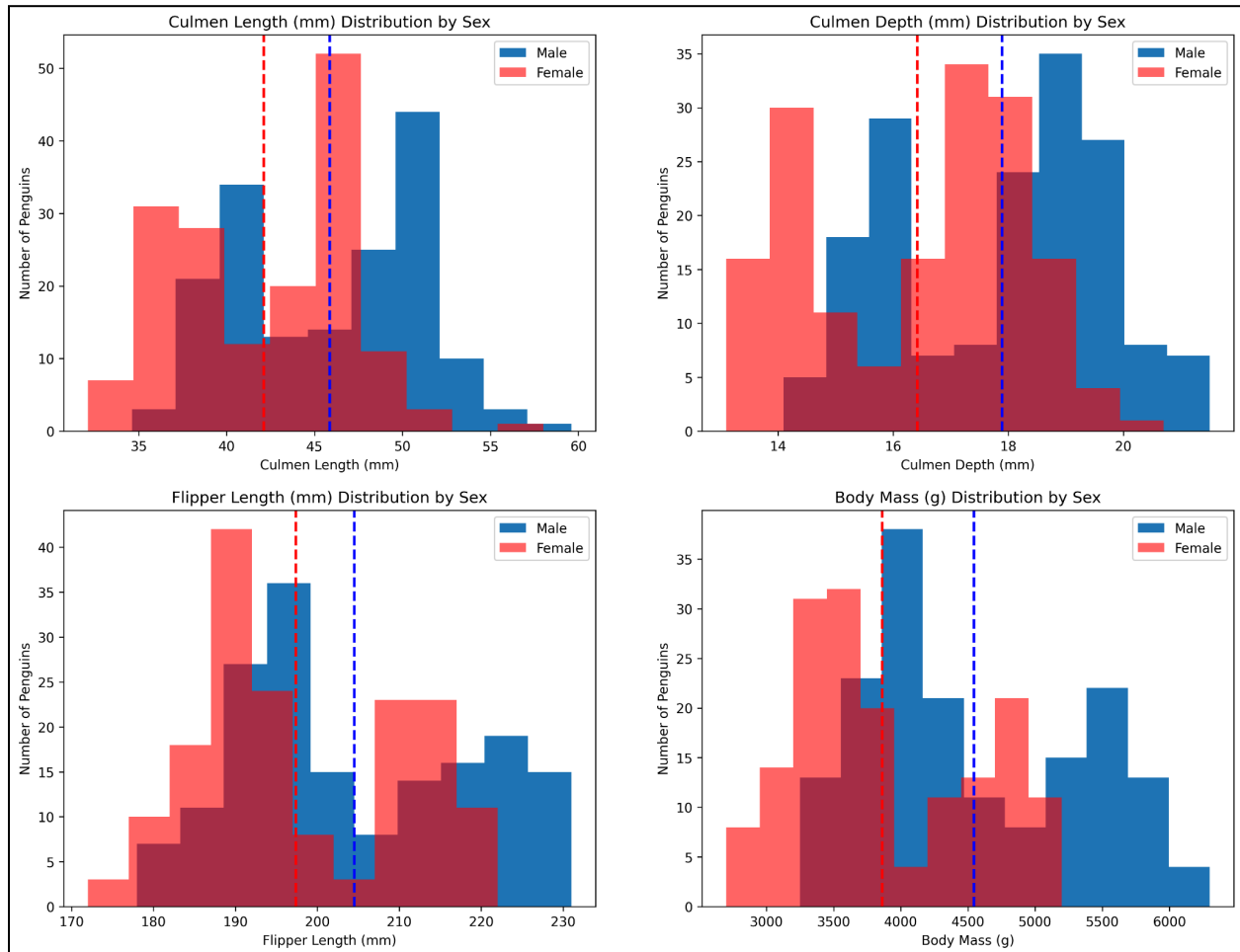
Figure 1:

Table of Summary Statistics for Mean Physical Measurements of Penguins by Sex

Sex	Culmen Length (mm)	Culmen Depth (mm)	Flipper Length (mm)	Body Mass (g)
FEMALE	42.096970	16.425455	197.363636	3862.272727
MALE	45.854762	17.891071	204.505952	4545.684524

Figure 2:

Histogram Comparing Physical Trait Distributions Between Male and Female Penguins



By calculating some summary statistics of the data, we found a number of points. For one, there are 10 rows in the data that have null values in the “Sex” column, so we wouldn’t be able to use those rows. That means that although there are 344 total samples, we would only be able to analyze the results from 334 of the samples. Later, when creating the predictive model, we found that there was a row in the “Sex” column that had “.” as its value, rather than “Male” or “Female”. After removing this, we were left with 333 samples to analyze.

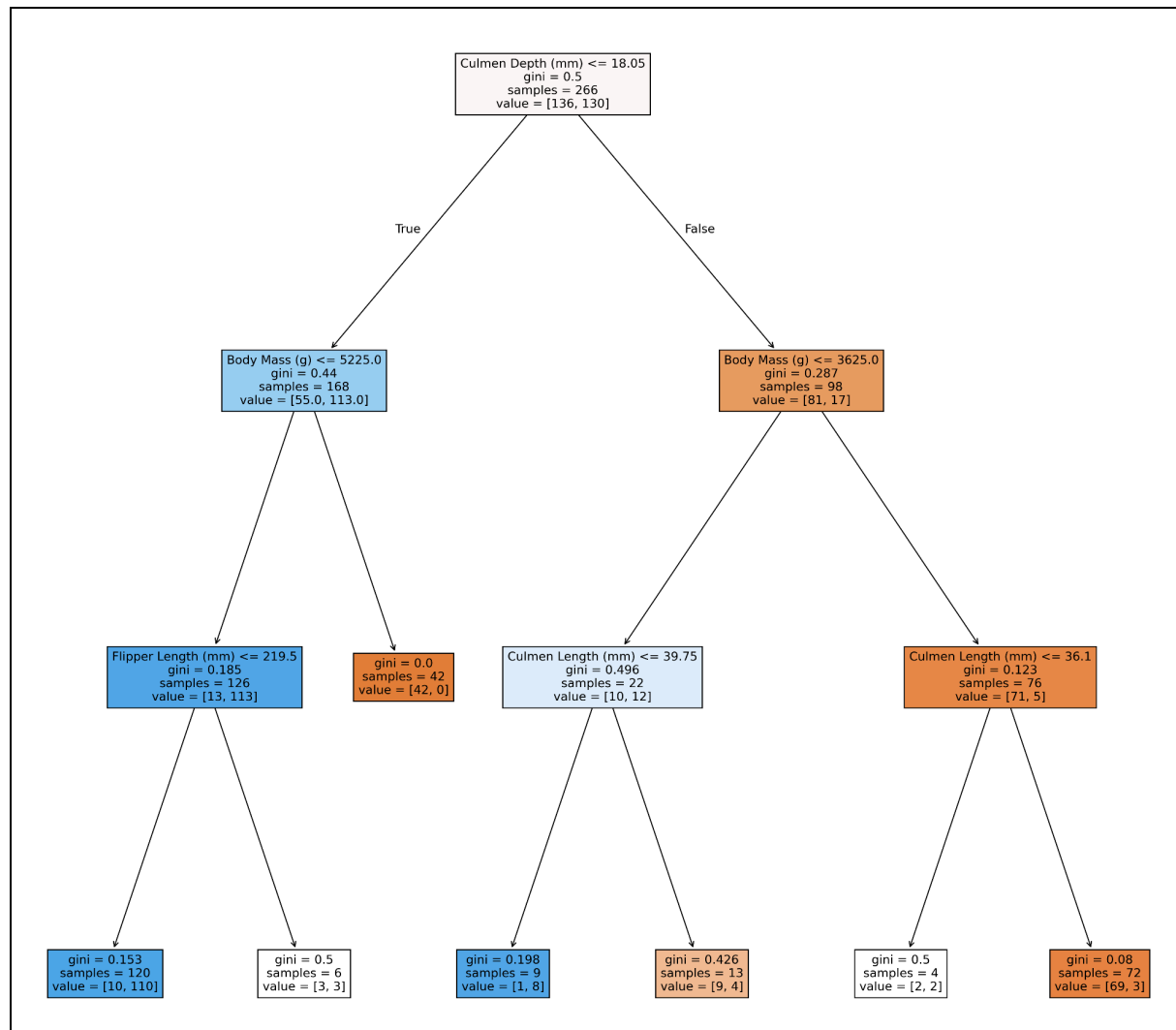
We also found that female penguins tend to weigh less than male penguins, with the average weights being ~3862 grams and ~4545 grams respectively. In addition, males also have longer flipper lengths on average than females (204.5 mm compared to 197.4 mm). On top of that, the average male penguin culmen length and depth are both longer than those of the female penguins. These patterns show that there are clear physical differences between the male and female penguins in the dataset, which means that there is a solid chance that we'll be able to accurately predict penguin sex from their physical characteristics. At the same time though, there is some overlap between the male and female penguins, especially over different species, so the model may not be able to perfectly predict the penguins' sex.

Model building

For our problem, we decided to build a Decision Tree Classifier. We chose this predictive model for two reasons. For one, our target variable was the sex of the penguin, which has the categorical values "Male" and "Female", which makes it a classification problem. Since a Decision Tree Classifier is a classification model, this makes it much more suitable for answering our question compared to a regression model. The second reason is that decision trees are easy to read and interpret, which is extremely important in our case. The tree allows us and the readers to see exactly which physical traits the model thinks are the most important at each split. Since part of our question asks which traits are the strongest indicators in predicting penguin sex, being able to see each split clearly is vital in answering the question.

Figure 3:

Image of Decision Tree Classifier for Predicting Penguin Sex



The dataset we used contains several features that we could have used, but we settled on four physical characteristics of the penguins in particular: culmen length (mm), culmen depth (mm), flipper length (mm), and body mass (g). We chose these four features because they represent measurable physical traits that could naturally differ between male and female penguins. They are also some of the most straightforward features in the dataset, making them easy to work with and interpret. From our data exploration we saw clear differences in the physical characteristics between the two sexes, with males generally having greater culmen length, culmen depth, flipper length, and body mass than females. We saw these differences in

both the summary statistics and in the histogram visualizations, making these features strong indicators for predicting penguins' sex.

Model evaluation

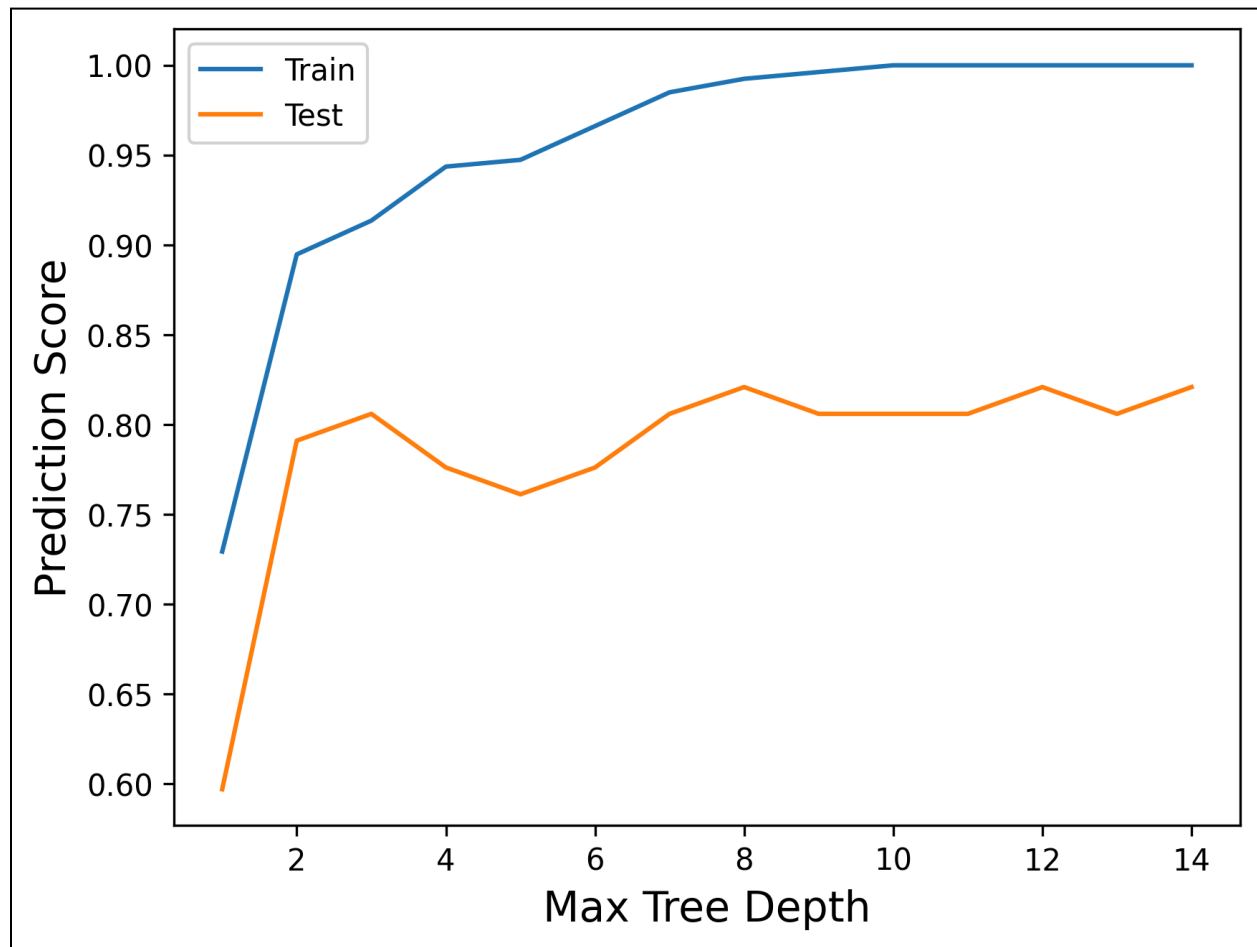
Again, our question asked if we can predict a penguin's sex based on its physical characteristics, and which traits are the strongest indicators. The Decision Tree Classifier model that we used provided us with a clear answer: physical characteristics of the penguin do have predictive capability. To evaluate the model, we used a standard train and test split, which is normal for evaluating a classification model. The model achieved an accuracy of 0.914 on the train set and an accuracy of 0.806 on the test set, meaning that it performs fairly well on new data and isn't overfitting or underfitting.

The tree first splits based on Culmen Depth (mm). The next split is based on Body Mass (g). Then the last split is based on Flipper Length (mm) and Culmen Length (mm). The root node splits based on the most informative feature, which indicates Culmen Depth (mm) is the strongest indicator in determining penguins' sex. These splits also align with the differences we observed earlier, as male penguins tend to have deeper culmens, higher body mass, and longer flippers.

We know that our model is not overfitting because when testing max tree depths, we found the test score does not significantly increase past a max depth of 3. To find the "sweet spot" for max depth, we created a line plot that showcases the Decision Tree Classifier accuracy for the train and test data as max depth increases.

Figure 4:

Line Graph Depicting How Decision Tree Train and Test Accuracy Changes with Max Depth



As the graph shows, as we increase the max depth from 1 to 3, the train and test accuracy dramatically increases. When we increase from depth 3 to 5, the train accuracy increases, but the test accuracy decreases, showing signs of overfitting. However, when we increase the max depth past 5, the train accuracy continues to increase and the test accuracy increases as well. Although the test accuracy was improving, we decided to settle with a max depth of 3, as it has relatively the same accuracy score as higher max depths and the decision tree is much easier to read and interpret that way.

Our model does not have any other common issues that we might have run into. The model doesn't suffer from class imbalances as our sample size is fairly distributed, with the distribution being 168 male penguins and 165 female, so we didn't have to do any downsampling or upsampling. In addition, we decided not to normalize the features as decision trees split on cutoff values, not distances, so normalization would not have changed the model's performance. Overall, the evaluation shows that the model performs solidly, doesn't overfit, and makes decisions that actually line up with real biological differences in the physical traits.

Model deployment

A model like this could help researchers or wildlife biologists quickly estimate a penguin's sex when it isn't known or can't be observed directly or easily. It's also useful in early fieldwork, where taking a few simple measurements is much easier than doing any invasive investigation. Since the model relies on basic physical characteristics that are easy to collect, it can make data gathering easier and support studies where knowing the sex ratio matters.

As mentioned in the data collection section, one challenge is that the model is built on a specific dataset, so it might not generalize to penguins from different regions, species, or environments of Antarctica. Penguins' physical traits can change with age, health, or habitat, meaning these measurements alone could cause wrong predictions. Field measurements can also be inaccurate, and misreading culmen or flipper length would cause the models' accuracy to decrease. And importantly, our model should not replace previous biological knowledge or methods that might have been used to evaluate penguin sex. Machine-learning predictions can overlook or oversimplify real biological differences, and relying on just models could potentially lead to misleading or wrong conclusions.